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Editor-in-Chief
Quality and Reliability Engineering International

Dear Editor,

I am submitting the manuscript “When Does Model Simplification Matter? Consequence Analysis for Weibull Series Systems” for consideration as a research article in *Quality and Reliability Engineering International*.

This paper addresses a practical question in reliability engineering: when can a Weibull series system model be safely simplified by assuming all components share a common shape parameter? Through extensive Monte Carlo simulation under realistic conditions of right-censored and masked failure data, we demonstrate a favorable alignment between the prediction consequence of model misspecification and the statistical power to detect it. The common-shape model’s MTTF bias remains below 1% through shape CV $\approx 15\%$, while the likelihood ratio test lacks power precisely in this regime. We develop an adaptive LRT-based procedure that exploits this alignment, achieving RMSE within 2.5% of the full model at $n \geq 500$ while selecting the simpler model over 90% of the time when appropriate.

I believe this work is well suited for QREI given the journal’s emphasis on practical engineering aspects of quality and reliability, and its strong tradition in Weibull-based reliability methods and masked failure data analysis. The paper provides practitioners with both the assurance that a common simplification is safe and the tools to decide when it is not.

This manuscript has not been published or submitted elsewhere. The author declares no conflicts of interest.

Sincerely,

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